CSA5111 - Cryptography and Network Security

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CO1-AT1-APPLICATION BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classification encryption method. During testing a msg "MEET AT NONE" (1)

a. Reconstruct how the encrypt msg might appear if a shift 4 positions is applied

i. Encryption Rule:

Each plain text letter is shifted 4 positions forward alphabet.

ii. Encryption of all Letters:

Plain Text:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher Text:	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D

Plain Text: MEET AT NOON

Cipher Text: QIIX EX RSSR

∴ MEET AT NOON → QIIX EX RSSR

b Explain how an authorized member, aware of the encoding rule, would retrieve the original message

The authorized member uses the same Caesar cipher rule in reverse by shifting every cipher text letter four position backward

Cipher Text to Plain Text:

Cipher Text: QIIX EX RSSR

Key = 4

$P = (C-K) \% 26$

$C = Q$
$P = (16-4) \% 26$
$= 12 \% 26$
$= 12$
$= M$

$C = I$
$P = (8-4) \% 26$
$= 4 \% 26$
$= 4$
$= E$

$C = X$
$P = (23-4) \% 26$
$= 19 \% 26$
$= 19$
$= T$

$C = E$
$P = (4-4) \% 26$
$= 0 \% 26$
$= 0$
$= A$

$C = R$
$P = (17-4) \% 26$
$= (13) \% 26$
$= 13$
$= N$

$C = S$
$P = (18-4) \% 26$
$= 14 \% 26$
$= 14$
$= O$

$C = R \Rightarrow P = N$
$C = S \Rightarrow P = O$
$C = I \Rightarrow P = E$
$C = X \Rightarrow P = T$

∴ Plain Text: MEET AT NOON

2. Scenario: Intercepted Suspicious Message
A cybersecurity intern intercepts the coded message "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a. Deduce the most likely original message
This scenario uses Atbash cipher, a substitution ciphes in which the alphabet is reversed.

Atbash cipher:
The Atbash cipher substitutes each letter with its opposite in the alphabet
$A \rightarrow Z$, $B \rightarrow Y$, $C \rightarrow X$

Plain Text:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Cipher Text:	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L

	P	Q	R	S	T	U	V	W	X	Y	Z
	K	J	I	H	G	F	E	D	C	B	A

Cipher Text: GSRH RH Z HVXIVG

Plain Text: THIS IS A SECRET

∴ Plain Text: THIS IS A SECRET

b. Justify your answer by analysing repeating patterns, letter structures or known cipher characteristics

1. Reversed alphabet: Atbash maps each letter to its opposite in the alphabet. For example, G becomes T, S becomes H, R becomes I

2. Word structure: The decoded msg contains 4 words with lengths 4, 2, 1 and 6.

3. Meaningful Plaintext: The result, "THIS IS A SECRET", forms a grammatically meaningful English sentence.

4. Simple Substitution: Each ciphertext letter consistently maps to One plaintext letter, which is characteristic of a monoalpha-betic substitution cipher

∴ The message is most likely encrypted using the Atbash cipher

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using

the Keyword "KEY" to ensure confidentiality

a) Demonstrate how the encryption process transforms the msg step by step

This scenario uses the Vigenere cipher a polyalphabetic cipher substitution technique that encrypts a msg using a repeating keyword

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R

18	19	20	21	22	23	24	25
S	T	U	V	W	X	Y	Z

Plain Text: HELLO Keyword: KEY

H	E	L	L	O
K	E	Y	K	E

$C = (P_i + K_i) \% 26$

$P_i = H \quad K_i = K$

$C = (7 + 10) \% 26$

$= 17 \% 26$

$= 17$

$= R$

$P_i = E \quad K_i = E$

$C = (4 + 4) \% 26$

$= 8 \% 26$

$= 8$

$= I$

$P_i = L \quad K_i = Y$

$C = (11 + 24) \% 26$

$= 35 \% 26$

$= 9$

$= J$

$P_i = L \quad K_i = K$

$C = (11 + 10) \% 26$

$= 21 \% 26$

$= 21$

$= V$

$P_i = O \quad K_i = E$

$C = (14 + 4) \% 26$

$= 18 \% 26$

$= 18$

$= S$

∴ Cipher Text: RIJVS

b. Explain how the repeating keyword influences each character of

the Ciphertext

The repeating keyword determines the shifts applied to each plaintext letter. Each letter of keyword may produce a different shifts (3)

* K shifts H by 10 positions to R
* E shifts E by 4 positions to I
* Y shifts L by 24 positions to J, wrapping around the alphabet
* K repeats and shifts the next L to V
* E repeats and shifts O to S

Unlike the caesar cipher, which uses a fixed shift, the vigenere Ciphere uses multiple shifts based on the keyword. This makes simple frequency analysis more difficult, although short or reused keywords can still make it vulnerable to cryptanalysis

4 Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded with it The file appears normal upon casual inspection

a. Analyze how investigators could detect the presence of hidden data within the image

The scenario concerns steganography, the technique of hiding secret information inside an ordinary looking file

Investigators can use the following methods.

i. Metadata Analysis: Examine the img's metadata for unusual comments, unexpected fields or suspicious information.

ii. File Size Analysis: Compare the img size with similar imgs of the same dimensions and format. An unsually large files may warrnat further investigation.

iii. Steganography detection tools: Use tools such as stegExpose or other forensic utilities to identify possible hidden payloads or suspicious statistical patterns.

iv. Least Significant Bit (LSB) analysis: Examine pixel vals to identify unusual patterns that may indicate secret data encoded in the least significant bits

v. File signature and structure analysis: Inspect the file's internal structure for unexpected appended data, unusual chunks or inconsistencies blw file extension and its actual format.

vi. Comparison with the Original: If an original logo is available, Compare pixel vals and file structures to identify unexplained changes.

These methods provide evidence of possible hidden data, but no single test can reliably detect every steganographic technique.

b. Recommend two techniques that could strengthen the security of such hidden communications aganist detection or extraction.

i. Encrypt the secret data before embedding it.
Use a strong encryption alg such as AES-256 to encrypt the confidential msg before hiding it inside the img. If investigators extract the hidden data, they should not be able to understand it without the decryption key.

ii. Use a secure steganographic embedding method.
Use a method that minimizes detectable statistical changes to the image, rather than simply replacing predictable pixel bit. Apply appropriate embedding limits and avoid reusing the same embedding patterns accross multiple imgs.

5. Scenario : Legacy Encryption in corporate Communication (4)
A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received msg appear as: "WECRLTEERDSOEEFEAOCAIVDEN"

a. Determine the most probable original msg by reversing the encryption process

Cipher Text : "WECRL TEERDSOEEFEAOCAIVDEN

Rail-fence : 3

* For a three rail fence cipher, the plaintext is written in a zig-zag pattern across 3 rows.

Ciphertext contains 25 letters

Rail 1: positions 1,5, 9,13,17,21,25 → 7 letters

Rail 2: positions 2,4,6,8,10,12,14,16,18, 20,22,24 → 12 letters

Rail 3 : positions 3,7,11,15,19,23 → 6 letters

* In encryption the letters from Rail 1 are written first followed by Rail 2 and then Rail 3

W E C R L T E E
E R D S O E E F E A O C
A I V D E N

* Now take letters from the three groups in orders required by zigzag

Plain Text: "WEAREDISCOVEREDFLEEATONCE"

b. Critically evaluate why such a method is vunlerable in modern communication systems.

i. **No character substitution**: The ciphers only changes character positions, leaving the original letters unchanged.

ii. **Small key space**: The number of rails is usually small, so an attackes can try different rail counts.

iii. **Pattern Preservations**: The frequency of each letter remains exactly the same as in plaintext.

iv. **Brute-Force Attacks**: Short msg can be decrypted by testing several possible rail counts.

v. **Lack of modern security**: The method doesn't provide strong confidentiality aganist modern cryptanlysis & computing capabilities

c. Propose one practical enchancement to improve the confidentiality of msg in this scenario.

Replace the Rail Fence cipher with AES-256 encryption in an authenticated mode such as AES-GCM using securely generated keys and unique nonces

AES-GCM provides confidentiality and integrity protection, helping prevent unauthorized reading & undetected modification of mgs. The encryption key should be securely managed and never transmitted alongside the encrypted msg.